

# Benchmark comparison tables — acquisition $\times$ surrogate model

**Legend.** Each cell is the final metric (mean  $\pm$  sd over seeds). **Bold** marks the best model for that acquisition; **green** marks the single best (acquisition, model) combination for that problem. Lower is better for both metrics.

Table 1: Final regret (value  $- f^*$ ) for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 3$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                             | Branin                                 | Golinski                                     | Griewank 2-D                                                   | OTL Circuit                                                    |
|--------------------------|----------------------|----------------------------------------|----------------------------------------|----------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| LCB                      | Standard LVGP        | <b>0.00641 <math>\pm</math> 0.0063</b> | 0.0911 $\pm$ 0.057                     | $2.56 \times 10^{-6} \pm 8.5 \times 10^{-8}$ | <b>0.00413 <math>\pm</math> 0.015</b>                          | $1.53 \times 10^{-6} \pm 9.7 \times 10^{-7}$                   |
|                          | Heteroscedastic LVGP | 0.0302 $\pm$ 0.035                     | 0.0913 $\pm$ 0.057                     | $1.24 \times 10^{-7} \pm 1 \times 10^{-8}$   | 0.00858 $\pm$ 0.021                                            | <b>4.33 <math>\times 10^{-7} \pm 5.5 \times 10^{-7}</math></b> |
|                          | Separate GP          | 0.624 $\pm$ 1.2                        | 0.649 $\pm$ 1.1                        | <b>0 <math>\pm</math> 0</b>                  | 0.0912 $\pm$ 0.1                                               | 0.00142 $\pm$ 0.002                                            |
|                          | Categorical kernel   | 0.196 $\pm$ 0.71                       | <b>0.0516 <math>\pm</math> 0.088</b>   | <b>0 <math>\pm</math> 0</b>                  | 0.00441 $\pm$ 0.015                                            | $3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}$                   |
| PI                       | Standard LVGP        | <b>0.0127 <math>\pm</math> 0.015</b>   | <b>0.0697 <math>\pm</math> 0.057</b>   | 31.6 $\pm$ 17                                | <b>0.000736 <math>\pm</math> 0.0011</b>                        | 0.0242 $\pm$ 0.02                                              |
|                          | Heteroscedastic LVGP | 0.547 $\pm$ 1.2                        | 0.289 $\pm$ 0.91                       | 0.941 $\pm$ 1.9                              | 0.0395 $\pm$ 0.072                                             | 0.00164 $\pm$ 0.001                                            |
|                          | Separate GP          | 0.12 $\pm$ 0.55                        | 0.179 $\pm$ 0.42                       | <b>0 <math>\pm</math> 0</b>                  | 0.026 $\pm$ 0.048                                              | <b>1.74 <math>\times 10^{-5} \pm 2.3 \times 10^{-5}</math></b> |
|                          | Categorical kernel   | 0.0145 $\pm$ 0.012                     | 0.0736 $\pm$ 0.088                     | <b>0 <math>\pm</math> 0</b>                  | 0.000795 $\pm$ 0.0019                                          | 0.0155 $\pm$ 0.01                                              |
| EI                       | Standard LVGP        | <b>0.0123 <math>\pm</math> 0.012</b>   | <b>0.0532 <math>\pm</math> 0.054</b>   | 16.3 $\pm$ 17                                | <b>3.11 <math>\times 10^{-7} \pm 3.7 \times 10^{-6}</math></b> | 0.0565 $\pm$ 0.0                                               |
|                          | Heteroscedastic LVGP | 0.0212 $\pm$ 0.024                     | 0.092 $\pm$ 0.049                      | $0.000119 \pm 0.00015$                       | 0.00697 $\pm$ 0.019                                            | 0.0119 $\pm$ 0.01                                              |
|                          | Separate GP          | 0.221 $\pm$ 0.76                       | 0.419 $\pm$ 0.71                       | <b>0 <math>\pm</math> 0</b>                  | 0.0292 $\pm$ 0.049                                             | <b>0.00033 <math>\pm</math> 0.0004</b>                         |
|                          | Categorical kernel   | 0.0132 $\pm$ 0.014                     | 0.0898 $\pm$ 0.092                     | <b>0 <math>\pm</math> 0</b>                  | 0.000435 $\pm$ 0.00059                                         | 0.0218 $\pm$ 0.01                                              |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | 0.0166 $\pm$ 0.016                     | 0.0862 $\pm$ 0.054                     | 3.75 $\pm$ 2.1                               | 0.00464 $\pm$ 0.016                                            | 0.0268 $\pm$ 0.02                                              |
|                          | Separate GP          | 0.217 $\pm$ 0.77                       | 0.0966 $\pm$ 0.12                      | <b>0 <math>\pm</math> 0</b>                  | 0.0271 $\pm$ 0.049                                             | <b>0.000173 <math>\pm</math> 0.0002</b>                        |
|                          | Categorical kernel   | <b>0.0122 <math>\pm</math> 0.012</b>   | <b>0.0842 <math>\pm</math> 0.088</b>   | <b>0 <math>\pm</math> 0</b>                  | <b>0.00088 <math>\pm</math> 0.0015</b>                         | 0.0203 $\pm$ 0.01                                              |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | 0.024 $\pm$ 0.025                      | <b>0.078 <math>\pm</math> 0.085</b>    | 0.307 $\pm$ 0.61                             | <b>0.0137 <math>\pm</math> 0.024</b>                           | 0.0147 $\pm$ 0.01                                              |
|                          | Separate GP          | 0.027 $\pm$ 0.027                      | 0.19 $\pm$ 0.39                        | <b>0 <math>\pm</math> 0</b>                  | 0.0467 $\pm$ 0.061                                             | <b>0.000322 <math>\pm</math> 0.000</b>                         |
|                          | Categorical kernel   | <b>0.0191 <math>\pm</math> 0.024</b>   | 0.131 $\pm$ 0.41                       | 1.49 $\pm$ 3                                 | 0.0267 $\pm$ 0.029                                             | 0.0146 $\pm$ 0.01                                              |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b>0.0264 <math>\pm</math> 0.024</b>   | <b>0.00249 <math>\pm</math> 0.0073</b> | <b>9.65 <math>\pm</math> 6.6</b>             | 0.00853 $\pm$ 0.021                                            | <b>6.33 <math>\times 10^{-7} \pm 4.8 \times 10^{-7}</math></b> |
|                          | Separate GP          | 0.723 $\pm$ 1.3                        | 0.377 $\pm$ 0.91                       | 56.7 $\pm$ 47                                | 0.1 $\pm$ 0.12                                                 | 0.00142 $\pm$ 0.002                                            |
|                          | Categorical kernel   | 0.11 $\pm$ 0.55                        | 0.0269 $\pm$ 0.081                     | 20 $\pm$ 29                                  | <b>0.00463 <math>\pm</math> 0.015</b>                          | $1.23 \times 10^{-5} \pm 2.5 \times 10^{-6}$                   |

Table 2: Final regret (value  $-f^*$ ) for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 10$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                             | Branin                                 | Golinski                                     | Griewank 2-D                                                    | OTL Circ                                                       |
|--------------------------|----------------------|----------------------------------------|----------------------------------------|----------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| LCB                      | Standard LVGP        | <b>0.00408 <math>\pm</math> 0.0047</b> | <b>0.00124 <math>\pm</math> 0.0059</b> | $2.5 \times 10^{-6} \pm 8.5 \times 10^{-8}$  | <b>0.00413 <math>\pm</math> 0.015</b>                           | $3.87 \times 10^{-6} \pm 3.2 \times 10^{-7}$                   |
|                          | Heteroscedastic LVGP | $0.0188 \pm 0.021$                     | $0.0993 \pm 0.063$                     | $1.26 \times 10^{-7} \pm 1.3 \times 10^{-9}$ | $0.0105 \pm 0.023$                                              | $4.64 \times 10^{-7} \pm 3.8 \times 10^{-8}$                   |
|                          | Separate GP          | $0.726 \pm 1.3$                        | $0.529 \pm 0.88$                       | <b>0 <math>\pm</math> 0</b>                  | $0.0769 \pm 0.094$                                              | $0.00142 \pm 0.0001$                                           |
|                          | Categorical kernel   | $0.294 \pm 0.88$                       | $0.0941 \pm 0.12$                      | <b>0 <math>\pm</math> 0</b>                  | $0.0183 \pm 0.06$                                               | <b>0 <math>\pm</math> 0</b>                                    |
| PI                       | Standard LVGP        | $0.0131 \pm 0.014$                     | <b>0.00462 <math>\pm</math> 0.023</b>  | $43.8 \pm 0$                                 | <b>0.0009 <math>\pm</math> 0.0012</b>                           | $0.055 \pm 0.001$                                              |
|                          | Heteroscedastic LVGP | $0.28 \pm 1$                           | $0.238 \pm 0.59$                       | $1.37 \pm 2.6$                               | $0.0426 \pm 0.081$                                              | $0.00558 \pm 0.0001$                                           |
|                          | Separate GP          | $0.709 \pm 1.3$                        | $0.238 \pm 0.59$                       | <b>0 <math>\pm</math> 0</b>                  | $0.0349 \pm 0.062$                                              | <b><math>6.94 \times 10^{-5} \pm 7 \times 10^{-6}</math></b>   |
|                          | Categorical kernel   | <b>0.00491 <math>\pm</math> 0.006</b>  | $0.0548 \pm 0.066$                     | <b>0 <math>\pm</math> 0</b>                  | $0.0025 \pm 0.011$                                              | $0.0158 \pm 0.001$                                             |
| EI                       | Standard LVGP        | $0.0102 \pm 0.0098$                    | <b>0.00181 <math>\pm</math> 0.0076</b> | $37.3 \pm 13$                                | <b><math>-6.54 \times 10^{-7} \pm 4.2 \times 10^{-7}</math></b> | $0.0659 \pm 0.001$                                             |
|                          | Heteroscedastic LVGP | $0.022 \pm 0.031$                      | $0.0802 \pm 0.058$                     | $6.39 \times 10^{-5} \pm 7.9 \times 10^{-5}$ | $0.0225 \pm 0.048$                                              | $0.0121 \pm 0.001$                                             |
|                          | Separate GP          | $0.422 \pm 1$                          | $0.662 \pm 0.95$                       | <b>0 <math>\pm</math> 0</b>                  | $0.029 \pm 0.049$                                               | <b><math>3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}</math></b> |
|                          | Categorical kernel   | <b>0.01 <math>\pm</math> 0.0092</b>    | $0.158 \pm 0.39$                       | <b>0 <math>\pm</math> 0</b>                  | $0.000174 \pm 0.00025$                                          | $0.0163 \pm 0.001$                                             |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | $0.0259 \pm 0.023$                     | <b>0.0211 <math>\pm</math> 0.035</b>   | $7.82 \pm 4.1$                               | <b>0.00868 <math>\pm</math> 0.021</b>                           | $0.0117 \pm 0.001$                                             |
|                          | Separate GP          | $0.425 \pm 1$                          | $0.434 \pm 0.74$                       | <b>0 <math>\pm</math> 0</b>                  | $0.0288 \pm 0.049$                                              | <b><math>3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}</math></b> |
|                          | Categorical kernel   | <b>0.00801 <math>\pm</math> 0.0085</b> | $0.0421 \pm 0.048$                     | <b>0 <math>\pm</math> 0</b>                  | $0.0095 \pm 0.021$                                              | $0.0155 \pm 0.001$                                             |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | $0.0208 \pm 0.019$                     | <b>0.00541 <math>\pm</math> 0.014</b>  | $0.965 \pm 1.4$                              | <b>0.0194 <math>\pm</math> 0.028</b>                            | $0.0039 \pm 0.0001$                                            |
|                          | Separate GP          | $0.116 \pm 0.55$                       | $0.666 \pm 1.1$                        | <b>0 <math>\pm</math> 0</b>                  | $0.0413 \pm 0.048$                                              | <b>0.000162 <math>\pm</math> 0.0001</b>                        |
|                          | Categorical kernel   | <b>0.00552 <math>\pm</math> 0.0094</b> | $0.135 \pm 0.41$                       | <b>0 <math>\pm</math> 0</b>                  | $0.0406 \pm 0.049$                                              | $0.0287 \pm 0.001$                                             |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b>0.0203 <math>\pm</math> 0.025</b>   | <b>0.00189 <math>\pm</math> 0.0049</b> | <b>2.44 <math>\pm</math> 2.5</b>             | <b>0.0085 <math>\pm</math> 0.021</b>                            | $2.74 \times 10^{-7} \pm 1.7 \times 10^{-8}$                   |
|                          | Separate GP          | $0.723 \pm 1.3$                        | $0.288 \pm 0.78$                       | $17.5 \pm 28$                                | $0.0768 \pm 0.094$                                              | $0.00142 \pm 0.0001$                                           |
|                          | Categorical kernel   | $0.397 \pm 1$                          | $0.154 \pm 0.57$                       | $10.6 \pm 4$                                 | $0.0411 \pm 0.1$                                                | <b>0 <math>\pm</math> 0</b>                                    |

Table 3: Final noise  $\sigma^2$  at the incumbent design for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 3$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                                      | Branin                              | Golinski                                       | Griewank 2-D                           | OTL Circuit                                         |                                                      |
|--------------------------|----------------------|-------------------------------------------------|-------------------------------------|------------------------------------------------|----------------------------------------|-----------------------------------------------------|------------------------------------------------------|
| LCB                      | Standard LVGP        | <b><math>0.01 \pm 5.3 \times 10^{-8}</math></b> | $23.8 \pm 22$                       | $576 \pm 3.9 \times 10^{-8}$                   | $0.00608 \pm 0.0012$                   | $0.000194 \pm 5.3 \times 10^{-11}$                  | $0.000247 \pm 7.7 \times 10^{-11}$                   |
|                          | Heteroscedastic LVGP | $0.01 \pm 9 \times 10^{-7}$                     | $26.6 \pm 22$                       | $576 \pm 4.9 \times 10^{-9}$                   | <b><math>0.00576 \pm 0.0016</math></b> | $0.000194 \pm 1.7 \times 10^{-11}$                  | $0.000256 \pm 8.1 \times 10^{-11}$                   |
|                          | Separate GP          | $0.016 \pm 0.012$                               | $23.7 \pm 27$                       | <b><math>576 \pm 0</math></b>                  | $0.00892 \pm 0.0073$                   | $0.000651 \pm 0.00091$                              | <b><math>0.000183 \pm 5.8 \times 10^{-11}</math></b> |
|                          | Categorical kernel   | $0.0111 \pm 0.0054$                             | <b><math>10.8 \pm 19</math></b>     | <b><math>576 \pm 0</math></b>                  | $0.00608 \pm 0.0012$                   | <b><math>0.000194 \pm 0</math></b>                  | $0.000236 \pm 6.1 \times 10^{-11}$                   |
| PI                       | Standard LVGP        | $0.01 \pm 2.2 \times 10^{-7}$                   | <b><math>12 \pm 19</math></b>       | $597 \pm 19$                                   | $0.0064 \pm 2.3 \times 10^{-6}$        | $0.000753 \pm 0.00028$                              | <b><math>0.000196 \pm 5.1 \times 10^{-11}</math></b> |
|                          | Heteroscedastic LVGP | $0.0118 \pm 0.0059$                             | $24.7 \pm 24$                       | $576 \pm 4 \times 10^{-5}$                     | $0.0066 \pm 0.0048$                    | $0.000514 \pm 0.00032$                              | $0.000256 \pm 2.3 \times 10^{-11}$                   |
|                          | Separate GP          | $0.011 \pm 0.0054$                              | $12.9 \pm 19$                       | <b><math>576 \pm 0</math></b>                  | <b><math>0.00556 \pm 0.0034</math></b> | <b><math>0.000194 \pm 0</math></b>                  | $0.00021 \pm 4.1 \times 10^{-11}$                    |
|                          | Categorical kernel   | <b><math>0.01 \pm 1.7 \times 10^{-7}</math></b> | $15.9 \pm 21$                       | <b><math>576 \pm 0</math></b>                  | $0.0064 \pm 3.9 \times 10^{-6}$        | $0.000908 \pm 0.0008$                               | $0.000202 \pm 6.1 \times 10^{-11}$                   |
| EI                       | Standard LVGP        | <b><math>0.01 \pm 1.5 \times 10^{-7}</math></b> | <b><math>10.4 \pm 18</math></b>     | $584 \pm 11$                                   | $0.0064 \pm 7.5 \times 10^{-9}$        | $0.00079 \pm 0.0002$                                | $0.000175 \pm 3.1 \times 10^{-11}$                   |
|                          | Heteroscedastic LVGP | $0.01 \pm 5.4 \times 10^{-7}$                   | $23.4 \pm 22$                       | $576 \pm 4 \times 10^{-6}$                     | $0.00592 \pm 0.0014$                   | $0.000514 \pm 0.00032$                              | $0.000245 \pm 1.1 \times 10^{-11}$                   |
|                          | Separate GP          | $0.012 \pm 0.0075$                              | $14.2 \pm 18$                       | <b><math>576 \pm 0</math></b>                  | <b><math>0.00524 \pm 0.0035</math></b> | <b><math>0.000275 \pm 9.9 \times 10^{-5}</math></b> | $0.000209 \pm 5.1 \times 10^{-11}$                   |
|                          | Categorical kernel   | $0.01 \pm 1.8 \times 10^{-7}$                   | $17.3 \pm 21$                       | <b><math>576 \pm 0</math></b>                  | $0.0064 \pm 1.2 \times 10^{-6}$        | $0.000514 \pm 0.00032$                              | <b><math>0.000173 \pm 6.4 \times 10^{-11}</math></b> |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | $0.01 \pm 2.7 \times 10^{-7}$                   | $25 \pm 22$                         | $576 \pm 9.1 \times 10^{-6}$                   | $0.00608 \pm 0.0012$                   | $0.000512 \pm 0.00032$                              | <b><math>6.4 \times 10^{-11}</math></b>              |
|                          | Separate GP          | $0.012 \pm 0.0075$                              | $16.4 \pm 21$                       | <b><math>576 \pm 0</math></b>                  | <b><math>0.0054 \pm 0.0034</math></b>  | <b><math>0.000235 \pm 8.1 \times 10^{-5}</math></b> | $0.000211 \pm 3.1 \times 10^{-11}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 1.5 \times 10^{-7}</math></b> | <b><math>14.9 \pm 21</math></b>     | <b><math>576 \pm 0</math></b>                  | $0.0064 \pm 3.1 \times 10^{-6}$        | $0.000275 \pm 0.0001$                               | $0.000185 \pm 6.1 \times 10^{-11}$                   |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | $0.01 \pm 4.8 \times 10^{-7}$                   | $15.1 \pm 21$                       | $576 \pm 0.00033$                              | $0.00544 \pm 0.0019$                   | $0.000275 \pm 9.9 \times 10^{-5}$                   | <b><math>5.89 \times 10^{-11}</math></b>             |
|                          | Separate GP          | $0.01 \pm 6.2 \times 10^{-7}$                   | $13.6 \pm 21$                       | <b><math>576 \pm 0</math></b>                  | $0.00505 \pm 0.0045$                   | <b><math>0.000275 \pm 9.9 \times 10^{-5}</math></b> | $0.000161 \pm 7.1 \times 10^{-11}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 6 \times 10^{-7}</math></b>   | <b><math>9.12 \pm 17</math></b>     | <b><math>576 \pm 0</math></b>                  | <b><math>0.00465 \pm 0.0023</math></b> | $0.000333 \pm 0.00028$                              | $0.000153 \pm 5.1 \times 10^{-11}$                   |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b><math>0.01 \pm 5.1 \times 10^{-7}</math></b> | <b><math>1.41 \pm 0.0085</math></b> | <b><math>576 \pm 2.9 \times 10^{-6}</math></b> | <b><math>0.00576 \pm 0.0016</math></b> | $0.000194 \pm 2.8 \times 10^{-11}$                  | $0.000194 \pm 2.8 \times 10^{-11}$                   |
|                          | Separate GP          | $0.017 \pm 0.013$                               | $7.48 \pm 18$                       | $592 \pm 26$                                   | $0.00839 \pm 0.0071$                   | $0.000651 \pm 0.00091$                              | <b><math>0.000198 \pm 5.8 \times 10^{-11}</math></b> |
|                          | Categorical kernel   | $0.011 \pm 0.0054$                              | $4.5 \pm 12$                        | $587 \pm 22$                                   | $0.00608 \pm 0.0012$                   | <b><math>0.000194 \pm 0</math></b>                  | $0.000231 \pm 3.1 \times 10^{-11}$                   |

Table 4: Final noise  $\sigma^2$  at the incumbent design for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 10$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                                      | Branin                              | Golinski                      | Griewank 2-D                           | OTL Circuit                                         |                                                     |
|--------------------------|----------------------|-------------------------------------------------|-------------------------------------|-------------------------------|----------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| LCB                      | Standard LVGP        | <b><math>0.01 \pm 2.6 \times 10^{-8}</math></b> | <b><math>1.41 \pm 0.0058</math></b> | $576 \pm 3.9 \times 10^{-8}$  | $0.00608 \pm 0.0012$                   | $0.000194 \pm 1.4 \times 10^{-10}$                  | $0.000256 \pm 7.3 \times 10^{-5}$                   |
|                          | Heteroscedastic LVGP | $0.01 \pm 4.1 \times 10^{-7}$                   | $27.9 \pm 22$                       | $576 \pm 5.2 \times 10^{-10}$ | <b><math>0.0056 \pm 0.0018</math></b>  | $0.000194 \pm 2.7 \times 10^{-11}$                  | $0.000256 \pm 1.7 \times 10^{-4}$                   |
|                          | Separate GP          | $0.017 \pm 0.013$                               | $22.8 \pm 24$                       | <b><math>576 \pm 0</math></b> | $0.00788 \pm 0.0068$                   | $0.000651 \pm 0.00091$                              | <b><math>0.000195 \pm 5.7 \times 10^{-5}</math></b> |
|                          | Categorical kernel   | $0.0121 \pm 0.0075$                             | $20.2 \pm 22$                       | <b><math>576 \pm 0</math></b> | $0.00712 \pm 0.0034$                   | <b><math>0.000194 \pm 0</math></b>                  | $0.000234 \pm 5.1 \times 10^{-5}$                   |
| PI                       | Standard LVGP        | $0.01 \pm 1.9 \times 10^{-7}$                   | <b><math>2.89 \pm 8</math></b>      | $605 \pm 0$                   | $0.0064 \pm 2.5 \times 10^{-6}$        | $0.00121 \pm 0.00064$                               | $0.000211 \pm 3.9 \times 10^{-5}$                   |
|                          | Heteroscedastic LVGP | $0.0115 \pm 0.0059$                             | $16 \pm 22$                         | $576 \pm 0.0015$              | $0.00736 \pm 0.0051$                   | $0.000474 \pm 0.00034$                              | $0.000256 \pm 5.3 \times 10^{-5}$                   |
|                          | Separate GP          | $0.0161 \pm 0.012$                              | $21.7 \pm 24$                       | <b><math>576 \pm 0</math></b> | <b><math>0.00584 \pm 0.0043</math></b> | <b><math>0.000194 \pm 0</math></b>                  | $0.000241 \pm 8.8 \times 10^{-5}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 4.3 \times 10^{-8}</math></b> | $11.4 \pm 18$                       | <b><math>576 \pm 0</math></b> | $0.00624 \pm 0.00086$                  | $0.000333 \pm 0.00028$                              | <b><math>0.000197 \pm 5.6 \times 10^{-5}</math></b> |
| EI                       | Standard LVGP        | $0.01 \pm 1.3 \times 10^{-7}$                   | <b><math>1.41 \pm 0.0072</math></b> | $602 \pm 6.2$                 | $0.0064 \pm 8.6 \times 10^{-10}$       | $0.000886 \pm 3.6 \times 10^{-6}$                   | <b><math>0.000159 \pm 5.3 \times 10^{-5}</math></b> |
|                          | Heteroscedastic LVGP | $0.01 \pm 9.3 \times 10^{-7}$                   | $20.9 \pm 22$                       | $576 \pm 3.9 \times 10^{-6}$  | $0.00572 \pm 0.0033$                   | $0.000474 \pm 0.00034$                              | $0.000233 \pm 4.7 \times 10^{-5}$                   |
|                          | Separate GP          | $0.014 \pm 0.01$                                | $25 \pm 26$                         | <b><math>576 \pm 0</math></b> | <b><math>0.00524 \pm 0.0035</math></b> | <b><math>0.000194 \pm 0</math></b>                  | $0.000241 \pm 8.8 \times 10^{-5}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 9.3 \times 10^{-8}</math></b> | $16.8 \pm 21$                       | <b><math>576 \pm 0</math></b> | $0.0064 \pm 5.1 \times 10^{-7}$        | $0.000235 \pm 8.1 \times 10^{-5}$                   | $0.000203 \pm 3.6 \times 10^{-5}$                   |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | $0.01 \pm 4.7 \times 10^{-7}$                   | <b><math>2.83 \pm 7.7</math></b>    | $576 \pm 7.1 \times 10^{-6}$  | $0.00576 \pm 0.0016$                   | $0.000474 \pm 0.00034$                              | <b><math>0.000124 \pm 5.3 \times 10^{-5}</math></b> |
|                          | Separate GP          | $0.014 \pm 0.01$                                | $21.8 \pm 23$                       | <b><math>576 \pm 0</math></b> | <b><math>0.00524 \pm 0.0035</math></b> | <b><math>0.000194 \pm 0</math></b>                  | $0.000225 \pm 3.8 \times 10^{-5}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 7.3 \times 10^{-8}</math></b> | $7.1 \pm 15$                        | <b><math>576 \pm 0</math></b> | $0.00576 \pm 0.0016$                   | $0.000413 \pm 0.00025$                              | $0.000142 \pm 6.3 \times 10^{-5}$                   |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | $0.01 \pm 3.2 \times 10^{-7}$                   | <b><math>1.41 \pm 0.013</math></b>  | $576 \pm 5.7 \times 10^{-6}$  | $0.00496 \pm 0.0022$                   | $0.000375 \pm 0.00027$                              | $0.000175 \pm 7.4 \times 10^{-5}$                   |
|                          | Separate GP          | $0.011 \pm 0.0054$                              | $12.1 \pm 22$                       | <b><math>576 \pm 0</math></b> | <b><math>0.00428 \pm 0.0037</math></b> | <b><math>0.000235 \pm 8.1 \times 10^{-5}</math></b> | $0.000199 \pm 4.4 \times 10^{-5}$                   |
|                          | Categorical kernel   | <b><math>0.01 \pm 9 \times 10^{-8}</math></b>   | $4.64 \pm 9.8$                      | <b><math>576 \pm 0</math></b> | $0.00445 \pm 0.0037$                   | $0.000377 \pm 0.00027$                              | <b><math>0.000137 \pm 2.7 \times 10^{-5}</math></b> |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b><math>0.01 \pm 5.5 \times 10^{-7}</math></b> | <b><math>1.41 \pm 0.0073</math></b> | $576 \pm 0.74$                | $0.00576 \pm 0.0016$                   | $0.000194 \pm 1.1 \times 10^{-11}$                  | $0.000256 \pm 6.6 \times 10^{-5}$                   |
|                          | Separate GP          | $0.017 \pm 0.013$                               | $3.41 \pm 9.6$                      | $589 \pm 25$                  | $0.00788 \pm 0.0068$                   | $0.000651 \pm 0.00091$                              | <b><math>0.000178 \pm 5.2 \times 10^{-5}</math></b> |
|                          | Categorical kernel   | $0.0131 \pm 0.009$                              | $2.34 \pm 4.7$                      | <b><math>576 \pm 0</math></b> | <b><math>0.00563 \pm 0.0033</math></b> | <b><math>0.000194 \pm 0</math></b>                  | $0.000236 \pm 6.2 \times 10^{-5}$                   |

Table 5: Final mean-variance regret (RAHBO robustness,  $MV = f + \alpha\sigma^2$ ,  $\alpha = 1$ ) for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 3$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                             | Branin                                 | Golinski                                     | Griewank 2-D                                                   | OTL Circui                                                     |
|--------------------------|----------------------|----------------------------------------|----------------------------------------|----------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
| LCB                      | Standard LVGP        | <b>0.00641 <math>\pm</math> 0.0063</b> | <b>0.344 <math>\pm</math> 0.54</b>     | $3.58 \times 10^{-6} \pm 1 \times 10^{-7}$   | <b>0.00381 <math>\pm</math> 0.014</b>                          | $1.53 \times 10^{-6} \pm 9.7 \times 10^{-7}$                   |
|                          | Heteroscedastic LVGP | $0.0302 \pm 0.035$                     | $0.793 \pm 1.1$                        | $1.71 \times 10^{-7} \pm 1.4 \times 10^{-8}$ | $0.00794 \pm 0.019$                                            | <b><math>4.33 \times 10^{-7} \pm 5.5 \times 10^{-7}</math></b> |
|                          | Separate GP          | $0.63 \pm 1.2$                         | $2.09 \pm 3.3$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0937 \pm 0.11$                                              | $0.00188 \pm 0.003$                                            |
|                          | Categorical kernel   | $0.197 \pm 0.71$                       | $1.11 \pm 3.9$                         | <b>0 <math>\pm</math> 0</b>                  | $0.00409 \pm 0.014$                                            | $3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}$                   |
| PI                       | Standard LVGP        | <b>0.0127 <math>\pm</math> 0.015</b>   | $0.571 \pm 1.4$                        | $48.7 \pm 30$                                | <b>0.000738 <math>\pm</math> 0.0011</b>                        | $0.0248 \pm 0.02$                                              |
|                          | Heteroscedastic LVGP | $0.549 \pm 1.2$                        | $2.53 \pm 4.5$                         | $0.941 \pm 1.9$                              | $0.0397 \pm 0.075$                                             | $0.00196 \pm 0.001$                                            |
|                          | Separate GP          | $0.121 \pm 0.56$                       | $0.736 \pm 1.5$                        | <b>0 <math>\pm</math> 0</b>                  | $0.0251 \pm 0.049$                                             | <b><math>1.74 \times 10^{-5} \pm 2.3 \times 10^{-5}</math></b> |
|                          | Categorical kernel   | $0.0145 \pm 0.012$                     | <b>0.273 <math>\pm</math> 0.46</b>     | <b>0 <math>\pm</math> 0</b>                  | $0.000797 \pm 0.0019$                                          | $0.0162 \pm 0.01$                                              |
| EI                       | Standard LVGP        | <b>0.0123 <math>\pm</math> 0.012</b>   | <b>0.0884 <math>\pm</math> 0.13</b>    | $24.7 \pm 26$                                | <b><math>3.12 \times 10^{-7} \pm 3.7 \times 10^{-6}</math></b> | $0.0571 \pm 0.0$                                               |
|                          | Heteroscedastic LVGP | $0.0212 \pm 0.024$                     | $0.516 \pm 1$                          | $0.000123 \pm 0.00015$                       | $0.00649 \pm 0.017$                                            | $0.0122 \pm 0.01$                                              |
|                          | Separate GP          | $0.223 \pm 0.77$                       | $1.19 \pm 1.7$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0281 \pm 0.05$                                              | <b>0.000411 <math>\pm</math> 0.0005</b>                        |
|                          | Categorical kernel   | $0.0132 \pm 0.014$                     | $0.332 \pm 0.54$                       | <b>0 <math>\pm</math> 0</b>                  | $0.000436 \pm 0.00059$                                         | $0.0222 \pm 0.01$                                              |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | $0.0166 \pm 0.016$                     | $0.412 \pm 0.78$                       | $3.75 \pm 2.1$                               | $0.00432 \pm 0.014$                                            | $0.0272 \pm 0.0$                                               |
|                          | Separate GP          | $0.219 \pm 0.78$                       | $0.527 \pm 1.1$                        | <b>0 <math>\pm</math> 0</b>                  | $0.0261 \pm 0.05$                                              | <b>0.000213 <math>\pm</math> 0.0003</b>                        |
|                          | Categorical kernel   | <b>0.0122 <math>\pm</math> 0.012</b>   | <b>0.294 <math>\pm</math> 0.54</b>     | <b>0 <math>\pm</math> 0</b>                  | <b>0.000882 <math>\pm</math> 0.0015</b>                        | $0.0204 \pm 0.01$                                              |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | $0.024 \pm 0.025$                      | <b>0.424 <math>\pm</math> 0.88</b>     | $0.307 \pm 0.61$                             | <b>0.0127 <math>\pm</math> 0.022</b>                           | $0.0148 \pm 0.01$                                              |
|                          | Separate GP          | $0.027 \pm 0.027$                      | $0.523 \pm 1.1$                        | <b>0 <math>\pm</math> 0</b>                  | $0.0453 \pm 0.063$                                             | <b>0.000403 <math>\pm</math> 0.000</b>                         |
|                          | Categorical kernel   | <b>0.0191 <math>\pm</math> 0.024</b>   | $0.474 \pm 1.1$                        | $1.49 \pm 3$                                 | $0.025 \pm 0.026$                                              | $0.0147 \pm 0.01$                                              |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b>0.0264 <math>\pm</math> 0.024</b>   | <b>0.00144 <math>\pm</math> 0.0029</b> | <b>9.65 <math>\pm</math> 6.6</b>             | $0.00789 \pm 0.019$                                            | <b><math>6.33 \times 10^{-7} \pm 4.8 \times 10^{-7}</math></b> |
|                          | Separate GP          | $0.73 \pm 1.3$                         | $0.915 \pm 2.1$                        | $72.8 \pm 60$                                | $0.102 \pm 0.13$                                               | $0.00188 \pm 0.003$                                            |
|                          | Categorical kernel   | $0.111 \pm 0.56$                       | $0.288 \pm 1.1$                        | $29 \pm 47$                                  | <b>0.00431 <math>\pm</math> 0.014</b>                          | $1.23 \times 10^{-5} \pm 2.5 \times 10^{-6}$                   |

Table 6: Final mean-variance regret (RAHBO robustness,  $MV = f + \alpha\sigma^2$ ,  $\alpha = 1$ ) for every acquisition  $\times$  surrogate model ( $n_{\text{rep}} = 10$ , mean  $\pm$  sd over seeds).

| Acquisition              | Model                | Ackley 2-D                             | Branin                                 | Golinski                                     | Griewank 2-D                                                    | OTL Circu                                                      |
|--------------------------|----------------------|----------------------------------------|----------------------------------------|----------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| LCB                      | Standard LVGP        | <b>0.00408 <math>\pm</math> 0.0047</b> | <b>0.00407 <math>\pm</math> 0.0032</b> | $3.48 \times 10^{-6} \pm 1.1 \times 10^{-7}$ | <b>0.00381 <math>\pm</math> 0.014</b>                           | $3.87 \times 10^{-6} \pm 3.2 \times 10^{-7}$                   |
|                          | Heteroscedastic LVGP | $0.0188 \pm 0.021$                     | $1 \pm 1.6$                            | $1.72 \times 10^{-7} \pm 1.8 \times 10^{-9}$ | $0.0097 \pm 0.021$                                              | $4.64 \times 10^{-7} \pm 3.8 \times 10^{-8}$                   |
|                          | Separate GP          | $0.733 \pm 1.3$                        | $2.34 \pm 3.4$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0784 \pm 0.1$                                                | $0.00188 \pm 0.003$                                            |
|                          | Categorical kernel   | $0.297 \pm 0.88$                       | $1.77 \pm 4.3$                         | <b>0 <math>\pm</math> 0</b>                  | $0.019 \pm 0.064$                                               | <b>0 <math>\pm</math> 0</b>                                    |
| PI                       | Standard LVGP        | $0.0131 \pm 0.014$                     | <b>0.139 <math>\pm</math> 0.74</b>     | $72.5 \pm 0$                                 | <b>0.000902 <math>\pm</math> 0.0012</b>                         | $0.056 \pm 0.02$                                               |
|                          | Heteroscedastic LVGP | $0.281 \pm 1$                          | $2.41 \pm 4.5$                         | $1.37 \pm 2.6$                               | $0.0436 \pm 0.085$                                              | $0.00586 \pm 0.009$                                            |
|                          | Separate GP          | $0.715 \pm 1.3$                        | $1.72 \pm 2.2$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0343 \pm 0.064$                                              | <b><math>6.94 \times 10^{-5} \pm 7 \times 10^{-6}</math></b>   |
|                          | Categorical kernel   | <b>0.00491 <math>\pm</math> 0.006</b>  | $0.854 \pm 3.3$                        | <b>0 <math>\pm</math> 0</b>                  | $0.00234 \pm 0.01$                                              | $0.0159 \pm 0.01$                                              |
| EI                       | Standard LVGP        | $0.0102 \pm 0.0098$                    | <b>0.00449 <math>\pm</math> 0.0035</b> | $62.9 \pm 19$                                | <b><math>-6.55 \times 10^{-7} \pm 4.2 \times 10^{-7}</math></b> | $0.0666 \pm 0.0$                                               |
|                          | Heteroscedastic LVGP | $0.022 \pm 0.031$                      | $0.868 \pm 1.7$                        | $6.77 \times 10^{-5} \pm 8.2 \times 10^{-5}$ | $0.0218 \pm 0.049$                                              | $0.0124 \pm 0.01$                                              |
|                          | Separate GP          | $0.426 \pm 1.1$                        | $1.81 \pm 2$                           | <b>0 <math>\pm</math> 0</b>                  | $0.0278 \pm 0.05$                                               | <b><math>3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}</math></b> |
|                          | Categorical kernel   | <b>0.01 <math>\pm</math> 0.0092</b>    | $0.765 \pm 1.3$                        | <b>0 <math>\pm</math> 0</b>                  | $0.000174 \pm 0.00025$                                          | $0.0163 \pm 0.01$                                              |
| HAEI ( $\gamma = 0.5$ )  | Heteroscedastic LVGP | $0.0259 \pm 0.023$                     | $0.17 \pm 0.79$                        | $7.82 \pm 4.1$                               | <b>0.00804 <math>\pm</math> 0.019</b>                           | $0.0119 \pm 0.01$                                              |
|                          | Separate GP          | $0.429 \pm 1.1$                        | $1.29 \pm 1.8$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0277 \pm 0.05$                                               | <b><math>3.07 \times 10^{-5} \pm 6.1 \times 10^{-6}</math></b> |
|                          | Categorical kernel   | <b>0.00801 <math>\pm</math> 0.0085</b> | <b>0.0895 <math>\pm</math> 0.19</b>    | <b>0 <math>\pm</math> 0</b>                  | $0.00886 \pm 0.02$                                              | $0.0157 \pm 0.01$                                              |
| ANPEI ( $\beta = 0.8$ )  | Heteroscedastic LVGP | $0.0208 \pm 0.019$                     | <b>0.00777 <math>\pm</math> 0.022</b>  | $0.965 \pm 1.4$                              | <b>0.018 <math>\pm</math> 0.026</b>                             | $0.00408 \pm 0.007$                                            |
|                          | Separate GP          | $0.117 \pm 0.56$                       | $1.42 \pm 2.1$                         | <b>0 <math>\pm</math> 0</b>                  | $0.0392 \pm 0.049$                                              | <b>0.000202 <math>\pm</math> 0.0003</b>                        |
|                          | Categorical kernel   | <b>0.00552 <math>\pm</math> 0.0094</b> | $0.159 \pm 0.47$                       | <b>0 <math>\pm</math> 0</b>                  | $0.0387 \pm 0.05$                                               | $0.0289 \pm 0.01$                                              |
| RAHBO ( $\alpha = 0.5$ ) | Heteroscedastic LVGP | <b>0.0203 <math>\pm</math> 0.025</b>   | <b>0.00399 <math>\pm</math> 0.012</b>  | <b>2.72 <math>\pm</math> 2.4</b>             | <b>0.00786 <math>\pm</math> 0.019</b>                           | $2.74 \times 10^{-7} \pm 1.7 \times 10^{-8}$                   |
|                          | Separate GP          | $0.73 \pm 1.3$                         | $0.581 \pm 1.5$                        | $28.8 \pm 51$                                | $0.0783 \pm 0.1$                                                | $0.00188 \pm 0.003$                                            |
|                          | Categorical kernel   | $0.4 \pm 1$                            | $0.717 \pm 3.1$                        | $10.6 \pm 4$                                 | $0.0403 \pm 0.1$                                                | <b>0 <math>\pm</math> 0</b>                                    |